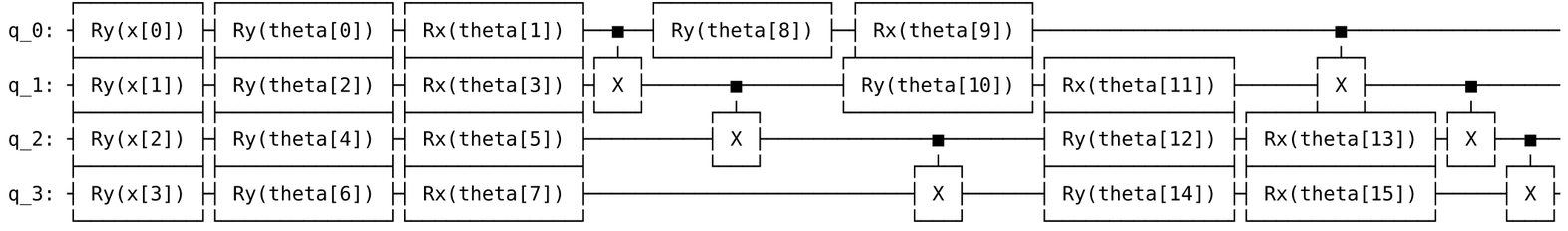
Four-qubit bottleneck: RY encoding + two RY/RX layers with directed CX chains



x[i] = pi tanh(h[i]); theta[0:16] trainable. Outputs: <Z0>, <Z1>, <Z2>, <Z3>. Ideal local simulation.